

Estimation for full regression testing of the learn.epam.com project

Step 1. Define key activities for full regression

Using the Work Breakdown Structure (WBS) technique

Phase	Subtasks
1. Environment setup	Install build, configure environment
2. Test case creation	Write 21 new test cases (Based on “Checklists and Test Cases” document)
3. Test case review & update	Update outdated ones (Based on “Checklists and Test Cases” document)
4. Regression test execution	Run all 29 test cases
5. Defect reporting & retesting	Log defects, verify fixes
6. Communication & reporting	Scrum ceremonies, daily reports

Step 2. Estimate each phase using different techniques

2.1 Environment setup - 3-Point Estimation

Formula: $ET = (O + 4M + P) / 6$

Scenario	Time (hours)	Notes
Optimistic (O)	1	If Sebastian really manages quickly
Most Likely (M)	4	Without David, setup may take longer
Pessimistic (P)	8	If configuration issues appear

$ET = (1 + 4 \cdot 4 + 8) / 6 = 4.17$ hours

Estimated Time: ~ 4.2 hours

2.2 Create 21 new test cases - Estimation by Test Cases Technique

In the “Checklists and Test Cases” document I have 29 tests in the checklist and 8 test cases, so I need to create 21 more.

Average creation time:

Small and simple: 10 min

Medium: 15 min

Complex and long: 20 min

Total = $21 \times 15\text{min} = 315\text{min} = 5.25\text{ hours}$

Add 15% buffer for review & clarifications:

$5.25 \times 1.15 = \sim 6\text{ hours}$

Estimated Time: 6 hours

2.3 Test case review & update

There are 29 tests in the checklist, of which ~25% need review/update.

$29 \times 25\% = 7\text{ tests to update}$

Estimate: 15 min per update - $7 \times 15\text{min} = 105\text{min}$

Add 30 min buffer for review - $135\text{min} = 2.25\text{h}$

Estimated Time: 2.25 hour

2.4 Full regression execution - Estimation by Test Cases Technique

Total: 29 test cases.

Complexity	N	Avg Time (min)	Subtotal (min)
Small and simple	15	5	75
Medium	10	10	100
Complex and long	4	15	60
Total (1 environment)	29	—	235 min = 3.9 h

Since it's full regression, tests must be repeated on 3 main environments (e.g., Chrome, Edge, Safari):

$3.9 \times 3 = 11.7\text{ hours}$

Add 15% buffer for bugs, environment instability, or re-runs:

$11.7 \times 1.15 = 13.5\text{ hours}$

Estimated Time: 13.5 hours

2.5 Defect reporting & retesting - 3-Point Estimation Technique

Formula: $ET = (O+4M+P) / 6$

Scenario	Time (hours)	Notes
Optimistic (O)	2	Few issues
Most Likely (M)	4	Normal defect flow
Pessimistic (P)	8	Multiple bug cycles

$ET = (2+4*4+8) / 6 = 4.33$ hours

Estimated Time: 4.33 hours

2.6 Communication & Reporting

- 0.5h stand-up × 5 days = 2.5h
- 0.5h daily report × 5 days = 2.5h

Estimated Time: 5 hours

Step 3. Combine Estimates

Task	Hours
Environment setup	4.2
Create 21 new test cases	6.0
Test case review & update	2.25
Full regression execution	13.5
Defect reporting & retesting	4.33
Communication & Reporting	5.0
Total (no buffer)	35.28 h

Full regression before release carries schedule risk (new build, missing DevOps support).

Add 20% contingency buffer: $35.28 * 1.2 = 42.3$ hours $\approx$ 5.3 working days

Estimation for full regression testing = 5.3 working days